

BOA JANG

Oculomics · Retinal Imaging Biomarkers · Longitudinal Biomarkers

I develop reliable retinal imaging biomarkers for systemic health and longitudinal disease monitoring.

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EDUCATION

Seoul National University <i>Ph.D. Candidate in Bioengineering</i>	Seoul, Korea Mar. 2023 – Present
Peking University <i>B.S. in Psychological and Cognitive Sciences</i>	Beijing, China Sep. 2016 – Jul. 2020

RESEARCH EXPERIENCE

Seoul National University Hospital (SNUH) <i>Medical AI Researcher</i> <i>Ph.D. Research Collaboration with SNUH</i>	Seoul, Korea Nov. 2021 – Feb. 2023 Mar. 2023 – Present
Developing AI for ophthalmic imaging in close collaboration with clinicians, spanning nAMD prognosis, retinal foundation models, and imaging biomarkers.	

Korea Institute of Science and Technology (KIST) <i>Research Intern, Brain Science Institute</i>	Seoul, Korea Aug. 2020 – May 2021
Studied basal ganglia circuits and motor control through in vivo optogenetics, behavioral experiments, and fluorescence imaging.	

MANUSCRIPTS UNDER REVIEW / IN REVISION

- [m.5] **B. Jang** et al. Retinal Age Gap as a Biomarker of Biological Aging Reflected by Socioeconomic Status and Lifestyle Behaviors. *GeroScience* (in revision).
- [m.4] **B. Jang** et al. Skeleton-Guided Progressive Test-Time Adaptation for Thin Curvilinear Structures. *AAAI 2027* (under review).
- [m.3] Integrating RNFL and Fundus Photographs for Differentiation of Glaucomatous and Non-Glaucomatous Optic Neuropathy Using Deep Learning. *Scientific Reports* (in revision).
- [m.2] Prediction of Postoperative Intracerebral Hemorrhage Following Brain Biopsy Using Integrated Statistical and Machine Learning Analyses. *Scientific Reports* (in revision).
- [m.1] Age- and Sex-Specific Autism Screening Using Machine Learning Reduction and Conversational Agent Integration. *Translational Psychiatry* (submitted).

PEER-REVIEWED PUBLICATIONS

[†] Equal contribution.

- [j.8] J.R. Song[†], **B. Jang**[†], et al. (2026). Segmentation-Guided Denoising Diffusion Probabilistic Models for Retinal OCT Speckle Noise Reduction with Enhanced Clinical Diagnostic Accuracy. *Scientific Reports* (Accepted).
- [j.7] **B. Jang**, Y. Ahn, et al. (2026). Clinically Grounded Retinal Representation Learning from Minimal Supervision. *Scientific Reports*.
- [j.6] **B. Jang**, Y. Shin, G. Lee, and Y.-G. Kim (2026). From Static Diagnosis to Dynamic Guidance: Evolution of Artificial Intelligence in Pediatric Neuroimaging. *Journal of Korean Neurosurgical Society*.
- [j.5] J.H. Lee, S.Y. Lee, **B. Jang**, et al. (2026). OCT Biomarkers as Predictors of Treatment Interval in Neovascular Age-Related Macular Degeneration Treated with Intravitreal Aflibercept Using a Treat-and-Extend Regimen. *Scientific Reports*.
- [j.4] **B. Jang**[†], C.H. Lee[†], et al. (2026). Artificial Intelligence-Based Prediction of First Recurrence in Neovascular Age-Related Macular Degeneration with Validation by 19 Experts. *Scientific Reports*.
- [j.3] D.Y. Kim, R. Do, . . . , **B. Jang**, et al. (2025). Automated AI-Based Identification of Autism Spectrum Disorder from Home Videos. *npj Digital Medicine*.
- [j.2] N.Y. Kim[†], **B. Jang**[†], et al. (2023). Differential Diagnosis of Pleural Effusion Using Machine Learning. *Annals of the American Thoracic Society*.
- [j.1] **B. Jang**[†], S.Y. Lee[†], et al. (2023). Preliminary Analysis of Predicting the First Recurrence in Patients with Neovascular Age-Related Macular Degeneration Using Deep Learning. *BMC Ophthalmology*.

PRESENTATIONS

- [o.7] **B. Jang**, G. Lee, J. Choi, and Y.-G. Kim. “Structure-Preserving Test-Time Adaptation for Cross-Modality Retinal Vessel Segmentation.” *2026 Spring Conference of The Korean Society of Medical Informatics (KoSMI)*, Cheongju, Korea.
- [po.3] **B. Jang**, K.H. Bae, and Y.-G. Kim. “Assessment of Systemic Health Using Retinal Age Gap: Development of a Predictive Model and Clinical Applicability.” *2025 Annual Conference of the Korea Society of Artificial Intelligence in Medicine (KoSAIM)*, Korea.
- [o.6] **B. Jang** et al. “Assessment of Systemic Health Using Retinal Age Gap: Development of a Predictive Model and Clinical Applicability.” *MICCAI 2025 AMAI Workshop*, Daejeon, Korea. **Best Abstract Award – Honorable Mention.**
- [o.5] **B. Jang** et al. “Assessment of Systemic Health Using Retinal Age Gap: Development of a Predictive Model and Clinical Applicability.” *2025 Spring Conference of The Korean Society of Medical Informatics (KoSMI)*, Busan, Korea. **Best Paper Award.**
- [o.4] **B. Jang** et al. “Self-Supervised Learning for Mesangial Proliferation Prediction in Digital Pathology.” *The 45th Annual Meeting of the Korean Society of Nephrology (KSN 2025)*, Seoul, Korea.
- [o.3] **B. Jang** and Y.-G. Kim. “Foundation Models in Healthcare: Development, Necessity and Implementation Strategies.” *2025 HIMSS Global Health Conference & Exhibition (HIMSS 2025)*, Las Vegas, NV, USA.
- [o.2] **B. Jang** et al. “The Impact of AI-Assistance on Predicting Recurrence in Neovascular Age-Related Macular Degeneration for Ophthalmologists.” *2023 Fall Conference of The Korean Society of Medical Informatics (KoSMI)*, Bundang, Korea. **Best Paper Award.**
- [po.2] **B. Jang** et al. “Development and Validation of Pretrained Model for Fundoscopy-Specific using Large-Scale Images.” *10th Workshop on Ophthalmic Medical Image Analysis (OMIA-X), 2023 in International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, Vancouver, Canada.
- [po.1] **B. Jang**, Y.-G. Kim, and E.K. Lee. “Predicting the First Recurrence in Patients with Neovascular AMD Using Deep Learning.” *2023 American Society of Retina Specialists (ASRS)*, Seattle, WA, USA. **Poster Award Winner.**
- [o.1] **B. Jang** et al. “Development and Validation of Pretrained Model for Fundoscopy-Specific using Large-Scale Images.” *2023 Spring Conference of The Korean Society of Medical Informatics (KoSMI)*, Daegu, Korea. **Best Paper Award.**

AWARDS

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|-------|---|------------|
| [a.5] | AI Seoul Tech Graduate Scholarship, Seoul Future Talent Foundation | Nov. 2025 |
| [a.4] | Hana Bank President’s Award, 2024 Digital Wellness Competition | Nov. 2024 |
| [a.3] | 2nd Place, Healthcare AI Learning Platform Challenge 2023, Themes 4 & 6 | Dec. 2023 |
| [a.2] | 3rd Place, Digital Pathology AI Hackathon, CODiPAI Challenge | Oct. 2023 |
| [a.1] | SNUAIMED Excellence Scholarship | 2023, 2024 |

TEACHING & MENTORING

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|-------|---|------------|
| [t.4] | Teaching Assistant, Korea Clinical Datathon (KCD) 2025 | Oct. 2025 |
| [t.3] | Student Ambassador, MICCAI 2025 AMAI Workshop | Sep. 2025 |
| [t.2] | Teaching Assistant, Clinical and Healthcare AI Training Programs | 2024, 2025 |
| [t.1] | Teaching Assistant, NVIDIA MONAI Bootcamp, HCLS Summit Korea 2022 | May 2022 |

PATENTS

- [p.2] Y.-G. Kim, . . . , **B. Jang**. Controllable Normal-to-Disease Fundus Translation via Factorial Guidance. Korea Patent Application No. 10-2026-0119988, filed June 2026. (*Pending*)
- [p.1] H.-S. Kim, . . . , Y.-G. Kim, **B. Jang**. Method and System for Predicting Information Related to Abdominal Surgery from Medical Image Data. Korea Patent No. 10-2953173, registered April 2026. (*Granted*)

REFERENCES

Prof. Jin-Wook Choi, PhD (jinchoi@snu.ac.kr)
Prof. Young-Gon Kim, PhD (younggon2.kim@gmail.com)
Prof. Hyuk Jin Choi, MD, PhD (docchoi@hanmail.net)